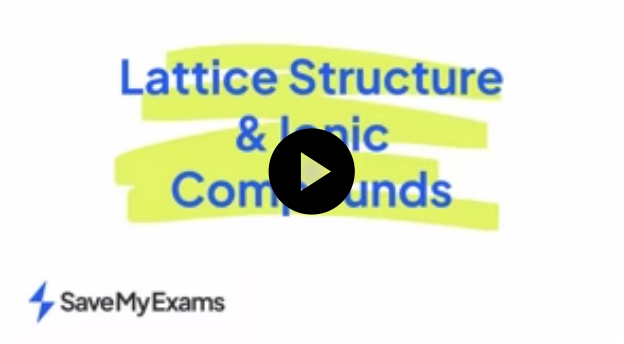
Ionic Bonds & Lattice Structure

Exam code: 0620 & 0971

Download PDF

Written by: Caroline Carroll
Reviewed by: Stewart Hird
Updated on 22 December 2025



Did this video help you? Yes No

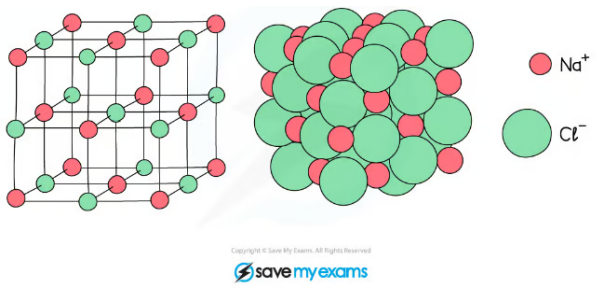
Help me understand

The lattice structure of ionic compounds

Extended Tier Only

Lattice structure

- Ionic compounds have a **giant lattice** structure
- Lattice structure refers to the arrangement of the atoms of a substance in 3D space
- In lattice structures, the atoms are arranged in an **ordered** and **repeating** fashion
- The lattices formed by ionic compounds consist of a **regular arrangement** of **alternating** positive and negative ions



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Ionic bonds between metallic & non-metallic elements

Extended Tier Only

Ionic compounds

- Ionic compounds are formed when metal atoms and non-metal atoms react
- The ionic compound has **no** overall charge

Example: magnesium Oxide, MgO

- Magnesium is a Group II metal so will lose two outer electrons to another atom to have a full outer shell of electrons
- A positive ion with the charge 2+ is formed
- Oxygen is a Group VI non-metal so will need to gain two electrons to have a full outer shell of electrons
- Two electrons will be transferred from the outer shell of the magnesium atom to the outer shell of the oxygen atom
- Oxygen atom will gain two electrons to form a negative ion with charge 2–
- Magnesium oxide has no overall charge
  - MgO

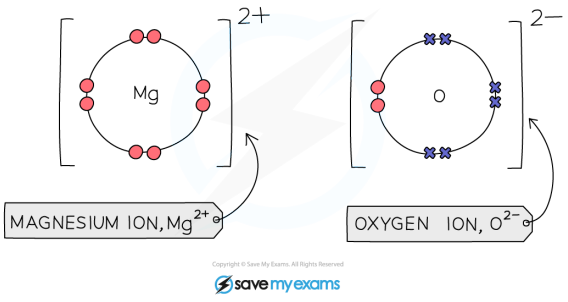


Diagram showing the dot-and-cross diagram of magnesium oxide

Examiner Tips and Tricks

When drawing dot and cross diagrams, you only need to show the outer shell of electrons. Remember to draw square brackets and include a charge for each ion. Make sure the overall charge is 0; you may need to include more than one positive or negative ion to ensure the positive and negative charges cancel each other out.

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Flashcards

Was this revision note helpful? Yes No

Previous

Next

Course

Study tools

All topics

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.